Further Consequences

Exact terminal contraction and the resulting upper bound

Dale James Raymond
ORCID: 0009-0001-1682-8529

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Companion to the *Version 1.0* of *A computer-assisted bound* $\Lambda \le 0.107964910079$ *for the de Bruijn–Newman constant.*
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Abstract

This note proves the consequences of a specified global terminal-strip estimate for the de Bruijn–Newman heat flow. It gives the full forward strip contraction, the exact rational completion time and the resulting two-sided enclosure after applying the non-negativity theorem of Rodgers and Tao. All displayed numerical comparisons follow from exact rational arithmetic. The terminal-strip estimate itself belongs to the main manuscript: this companion does not provide or certify its source proof.

This edition supersedes referee edition 3 of 28 October 2026; the numerical plan, source table and comparison history are unchanged; Appendix B adds the direct source reconstruction.

1 Normalisation and the precise input

Use the normalisation of Polymath [4],

$$H_t(z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du, \qquad H_0(z) = \tfrac{1}{8}\xi\Big(\tfrac{1+iz}{2}\Big), \qquad \partial_t H_t = -\partial_z^2 H_t.$$

Here Φ is the standard Riemann kernel and ξ is the completed Riemann xi function. The de Bruijn–Newman constant Λ is the threshold for all zeros of H_t to be real.

The input used by every proposition below is the following global assertion:

$$\begin{gathered} H_T(z) = 0 \quad \Longrightarrow \quad |\operatorname{Im} z|^2 \le Q_T, \\ T = \frac{22101}{204800}, \quad Q_T = \frac{7902377963830536747158643}{2^{96}}. \end{gathered} \tag{S}$$

The word *global* requires this statement for all zeros, without a finite restriction on $\operatorname{Re} z$. Establishing (S) from the source estimates, analytic comparison and finite computations is the task of the main manuscript [1]. The present deductions add no further strip hypothesis. Once (S) has been established unconditionally, the upper-bound conclusions below are unconditional as well.

2 Forward contraction and the exact endpoint

Proposition 1 (Global forward strips)**.** *Assume* (S)*. For every real $t \geq T$ and every zero z of H_t,*

$$|\operatorname{Im} z|^2 \leq \max\{Q_T - 2(t-T), 0\}.$$

Proof. In the stated heat-flow normalisation, the de Bruijn strip-contraction theorem [3, 4] sends a global strip with squared halfwidth q at time t_0 to a strip with squared halfwidth $\max\{q - 2(t-t_0), 0\}$ at every time $t \geq t_0$. Apply that theorem with $t_0 = T$ and $q = Q_T$. □

Corollary 2 (Real zeros at and beyond the exact completion time)**.** *Assume* (S)*. Set*

$$B_* := T + \frac{Q_T}{2} = \frac{427693072077942473661578098491}{396140812571321687967719751680 0}. \tag{1}$$

Every zero of H_t is real whenever $t \geq B_*$, and therefore

$$\Lambda \leq B_* < 0.107964910079 < 0.107965.$$

Proof. At $t = B_*$ the squared halfwidth in the proposition is zero; it remains zero for every later time. The threshold property of Λ gives $\Lambda \leq B_*$. The identity in (1) follows by putting T and $Q_T/2$ over the common denominator $25 \cdot 2^{97}$. The strict decimal comparisons are comparisons of rational numbers. □

Corollary 3 (Two-sided enclosure)**.** *Assume* (S)*. Then*

$$0 \leq \Lambda \leq B_* < 0.107964910079.$$

Proof. Combine the preceding corollary with the theorem $\Lambda \geq 0$ of Rodgers and Tao [5]. □

3 Exact numerical comparisons

The comparison time and the remaining contraction time are different quantities. In exact arithmetic,

$$T = 0.1079150390625, \qquad 0.0000498710162715 < \frac{Q_T}{2} < 0.0000498710162716.$$

The strict positivity of Q_T means that the terminal strip alone does not assert that all zeros are real at time T. The full contribution $Q_T/2$ must be retained in the stated endpoint. A useful decimal enclosure is

$$0.1079649100787715 < B_* < 0.1079649100787716.$$

The upper-bound display 0.107964910079 is a strictly larger rational number; the proof does not use rounded floating-point values.

For comparison with the earlier 0.116398 deposit [2], take the previously stated decimal ceiling as the exact comparison number

$$C_{\text{old}} = \frac{116397726408291}{10^{15}}.$$

Define $\Delta = C_{\text{old}} - B_*$. Then

$$0.008432816329519 < \Delta < 0.008432816329520,$$

and the relative reduction $100\Delta/C_{\mathrm{old}}$ is approximately 7.24483%. This is a comparison of the two stated numerical ceilings. It is not an estimate of the value of Λ, nor an independent proof of either ceiling's analytic premise.

The companion file `CONSEQUENCES_ARITHMETIC.json` records the exact fractions and each strict inequality. Its generator uses integer and rational arithmetic only. It checks these arithmetic consequences; it does not check (S) or execute the main proof packet.

4 Scope of the conclusions

The forward contraction above does not justify running the heat flow backwards to obtain a smaller time-zero zeta strip. It establishes neither $\Lambda < 0.1$ nor the Riemann or Lindelof hypotheses. The older consequence involving $H(34/43)$ and a sub-0.09 target requires an additional stronger strip statement and is not a consequence of (S).

For this review package, the main manuscript must establish (S) using its complete analytic argument and identified finite data. The source supplement must describe the source construction actually used in that manuscript, and the verification guide must match the corresponding executable packet. This note deliberately assigns no replay-stage count or scientific verification receipt to those separate materials.

References

[1] D. J. Raymond, *A computer-assisted bound $\Lambda \leq 0.107964910079$ for the de Bruijn–Newman constant*, version 1.0. Manuscript and proof materials, Zenodo record doi:10.5281/zenodo.23752103.

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[3] N. G. de Bruijn, *The roots of trigonometric integrals*, Duke Mathematical Journal **17** (1950), 197–226. doi:10.1215/S0012-7094-50-01720-0.

[4] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann ξ function, and a new upper bound for the de Bruijn–Newman constant*, Research in the Mathematical Sciences **6** (2019), article 31. doi:10.1007/s40687-019-0193-1.

[5] B. Rodgers and T. Tao, *The de Bruijn–Newman constant is non-negative*, Forum of Mathematics, Pi **8** (2020), e6. doi:10.1017/fmp.2020.6.